

Technical Note

CellularRAM™ Multiplexed Async/Burst Operation

Introduction

Micron CellularRAM™ products are high-speed CMOS PSRAM devices developed for low-power, portable applications. The popularity of the CellularRAM interface, as well as a continuing need for low pin-count, high-throughput PSRAM memory solutions, has prompted innovation resulting in the following implementation scheme.

The standard NOR Flash burst interface has been modified to allow address and data multiplexing on an MCP substrate or at the PCB level.

Figure 1 illustrates a 64Mb CellularRAM die placed on an MCP substrate. The CellularRAM die is the standard Micron async/page/burst device supporting separate address and data pins. The package substrate is used to connect A[15:0] to D[15:0] so that the controller “sees” A/DQ[15:0] as a multiplexed bus and A[21:16] as a normal (non-multiplexed) bus.

The CellularRAM device functions as per the current (non-multiplexed) data sheet with the exception of timing modifications (contained in this document) that allow A[15:0] and DQ[15:0] to share a common bus.

Figure 1: Substrate Illustrating AD Multiplexing

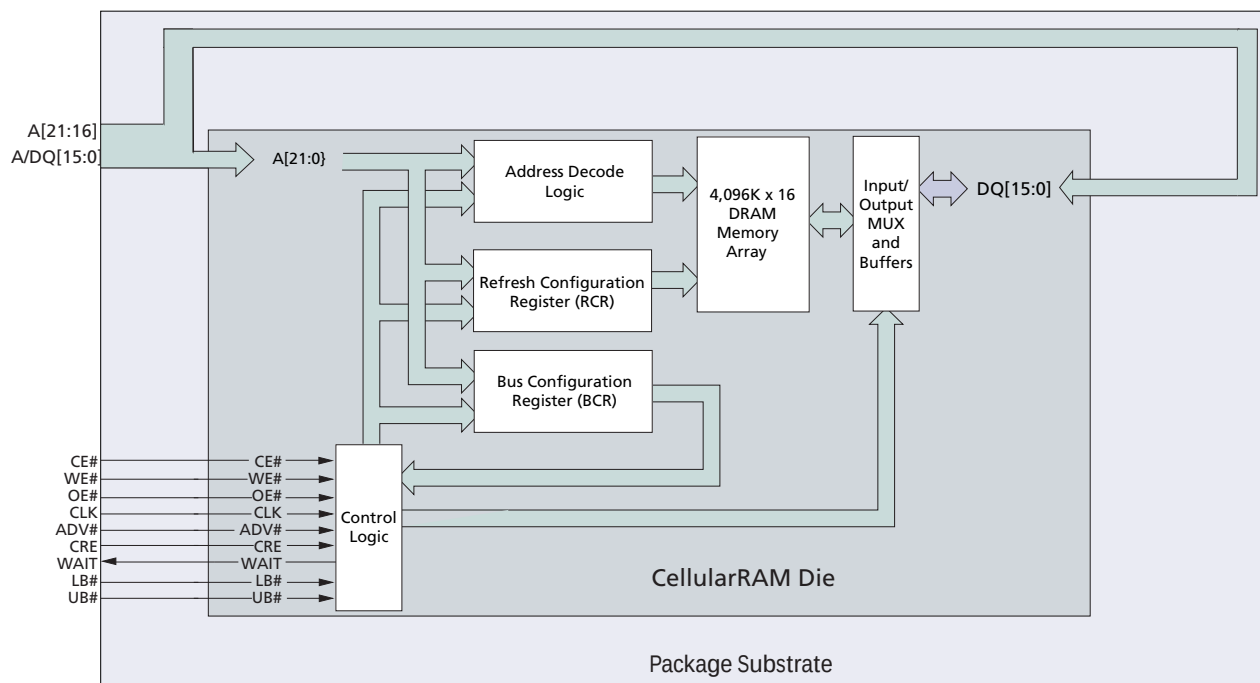


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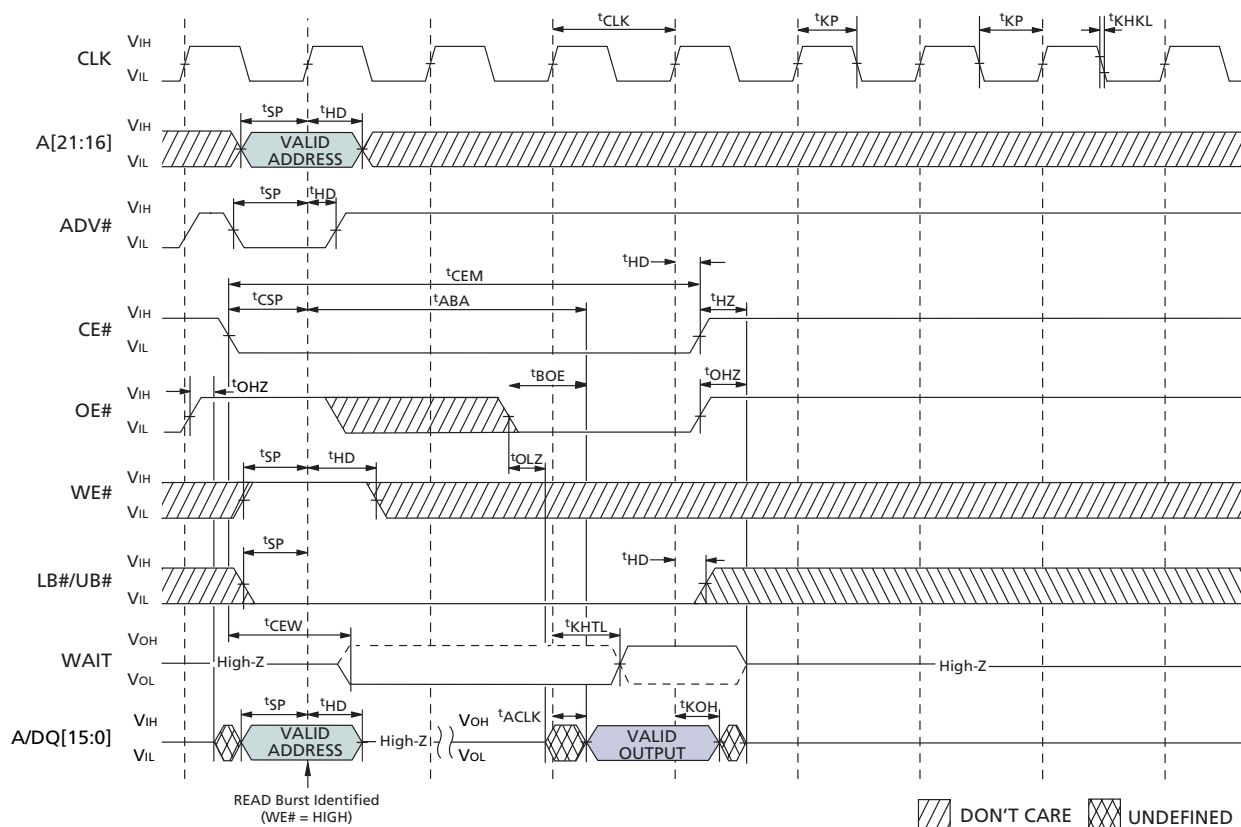
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The timing diagram for the AD9080 shows the following signals and parameters:

- Signals:** A[21:16], ADV#, CE#, LB#/UB#, OE#, WE#, A/DQ[15:0], WAIT.
- Timing Parameters:**
 - t_{VPH} : ADV# setup time before A[21:16] becomes valid.
 - t_{AVS} : Address setup time before ADV# becomes valid.
 - t_{AVH} : Address hold time after ADV# becomes valid.
 - t_{AA} : Address arrival time from A[21:16] to ADV#.
 - t_{AADV} : Address arrival time from A[21:16] to ADV#.
 - t_{TP} : ADV# setup time before CE# becomes valid.
 - t_{CVS} : CE# setup time before ADV# becomes valid.
 - t_{CO} : CE# setup time before LB#/UB# becomes valid.
 - t_{BA} : LB#/UB# setup time before ADV# becomes valid.
 - t_{OE} : OE# setup time before ADV# becomes valid.
 - t_{OH} : OE# hold time after ADV# becomes valid.
 - t_{BHZ} : LB#/UB# hold time after ADV# becomes valid.
 - t_{OHZ} : OE# hold time after ADV# becomes valid.
 - t_{BLZ} : WE# setup time before ADV# becomes valid.
 - t_{LZ} : WE# hold time after ADV# becomes valid.
 - t_{CEWS} : CE# setup time before ADV# becomes valid.
 - t_{HZ} : High-Z time after ADV# becomes valid.
 - t_{OV} : Output valid time after ADV# becomes valid.
- Valid Address and Valid Output:** The diagram shows the 'VALID ADDRESS' period for A/DQ[15:0] and the 'VALID OUTPUT' period for A/DQ[15:0].

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t AA		70		85	ns
^t AADV		70		85	ns
^t AVH	5		5		ns
^t AVS	10		10		ns
^t BA		70		85	ns
^t BHZ		8		8	ns
^t BLZ	10		10		ns
^t CEW	1	7.5	1	7.5	ns
^t CO		70		85	ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t CVS	10		10		ns
^t HZ		8		8	ns
^t LZ	10		10		ns
^t OE		20		20	ns
^t OHZ		8		8	ns
^t OLZ	5		5		ns
^t VP	10		10		ns
^t VPH	10		10		ns

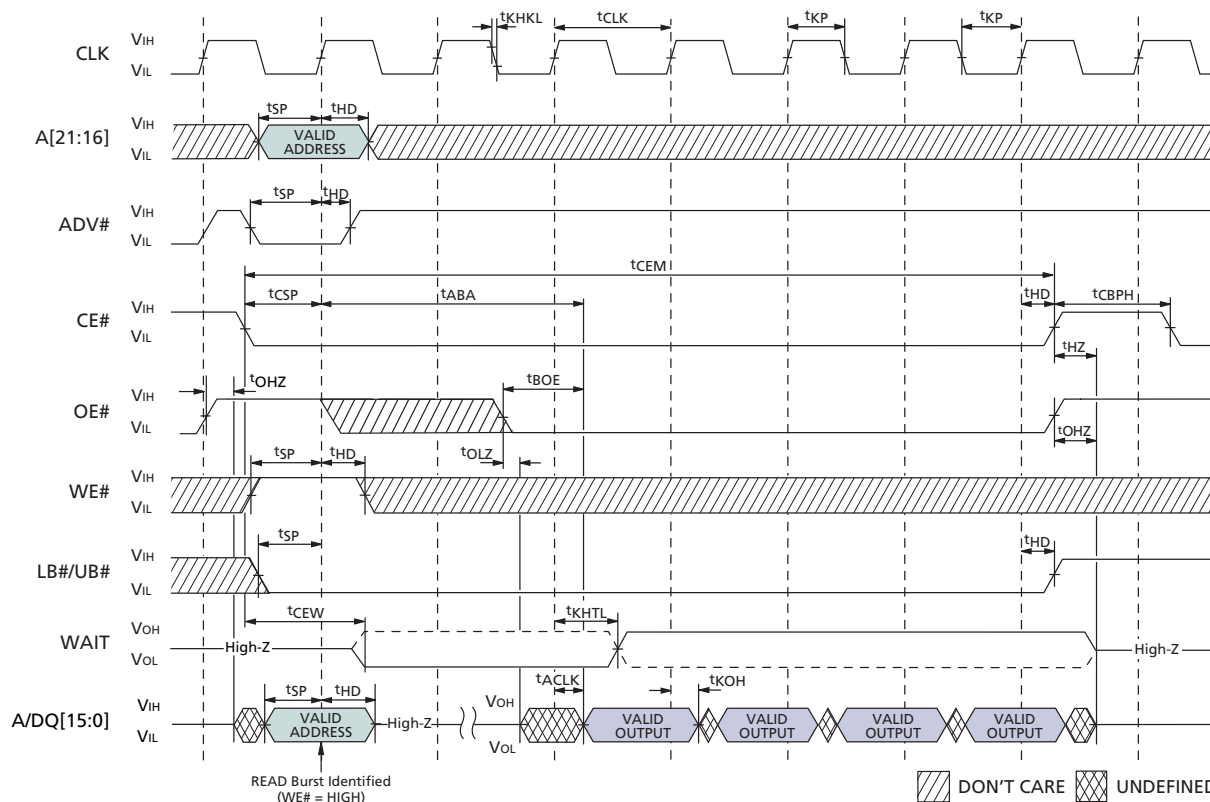
Figure 3: Single-Access Burst READ Operation


- Notes: 1. Non-default BCR settings for single-access burst READ operation: Latency code two (three clocks); WAIT active LOW; WAIT asserted during delay.
 2. Clock rates below 50 MHz ($t_{CLK} > 20\text{ns}$) are allowed as long as t_{CSP} specifications are met.

Table 2: Burst READ Timing Parameters

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{ABA}		46.5		56	ns
t_{ACK}		9		11	ns
t_{BOE}		20		20	ns
t_{CEM}		8		8	μs
t_{CEW}	1	7.5	1	7.5	ns
t_{CLK}	12.5	20	15	20	ns
t_{CSP}	4.5	20	5	20	ns
t_{HD}	2		2		ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{HZ}		8		8	ns
t_{KHKL}		1.8		2.0	ns
t_{KHTL}		9		11	ns
t_{KOH}	2		2		ns
t_{KP}	4		5		ns
t_{OHZ}		8		8	ns
t_{OLZ}	5		5		ns
t_{SP}	3		3		ns

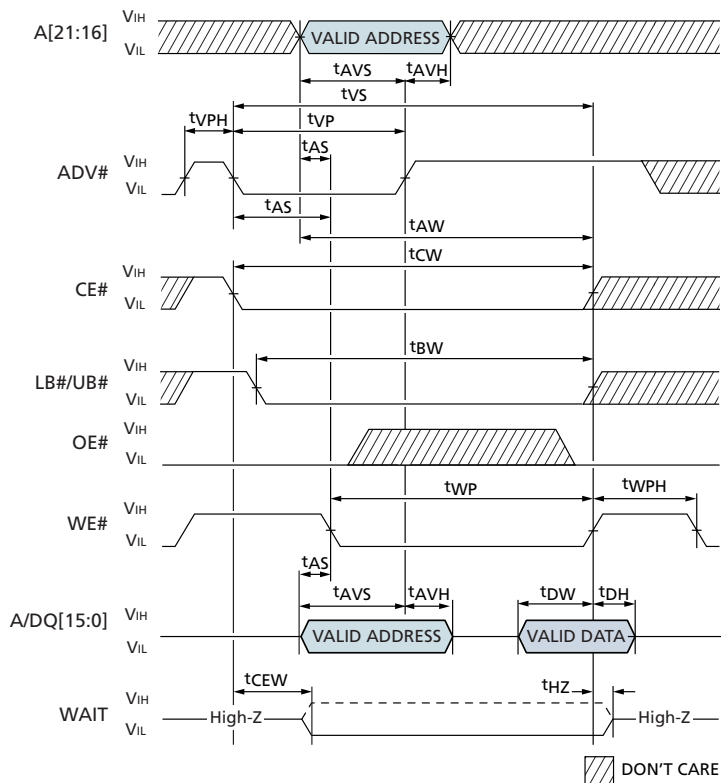
Figure 4: 4-Word Burst READ Operation


- Notes: 1. Non-default BCR settings for 4-word burst READ operation: Latency code two (three clocks); WAIT active LOW; WAIT asserted during delay.
 2. Clock rates below 50 MHz ($t_{CLK} > 20\text{ns}$) are allowed as long as t_{CSP} specifications are met.

Table 3: Burst READ Timing Parameters – 4-Word Burst

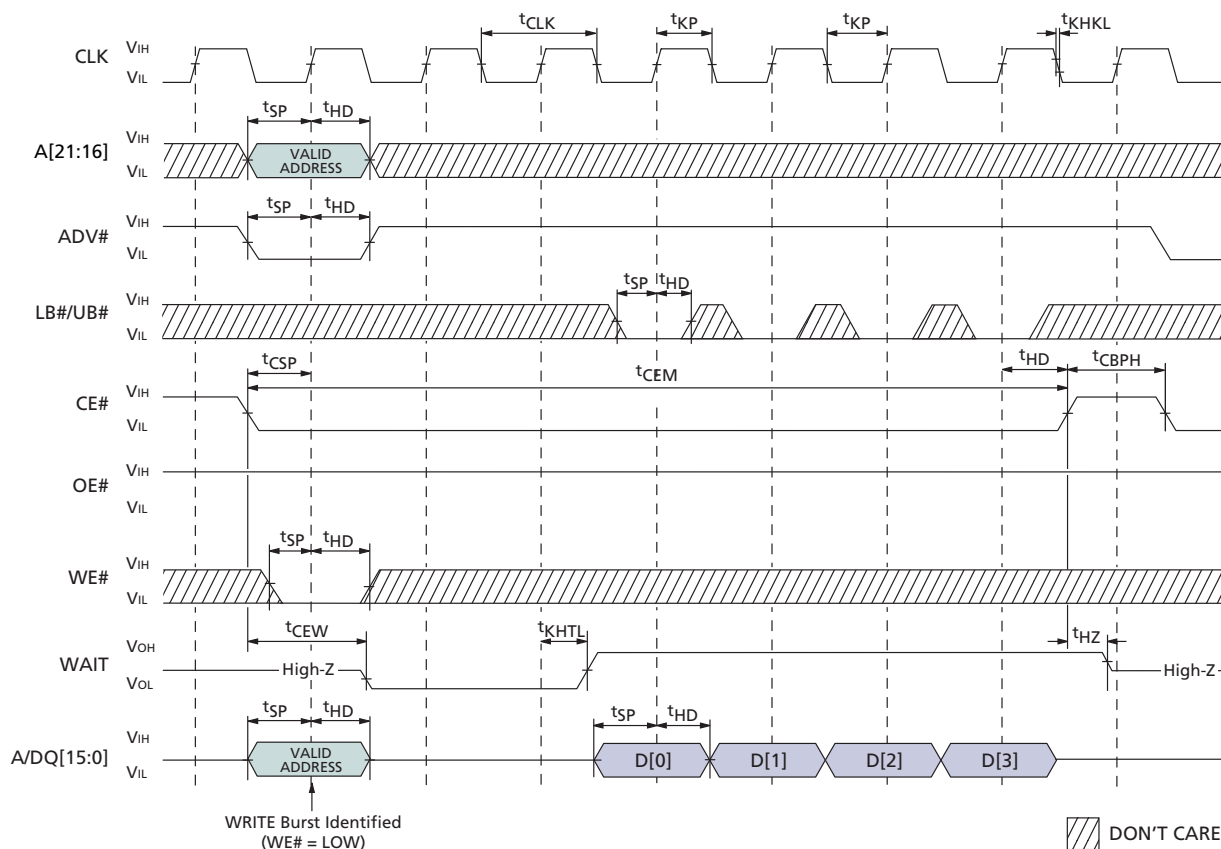
Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{ABA}		46.5		56	ns
t_{ACLK}		9		11	ns
t_{BOE}		20		20	ns
t_{CBPH}	5		5		ns
t_{CEM}		8		8	μs
t_{CEW}	1	7.5	1	7.5	ns
t_{CLK}	12.5	20	15	20	ns
t_{CSP}	4.5	20	5	20	ns
t_{HD}	2		2		ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{HZ}		8		8	ns
t_{KHKL}		1.8		2.0	ns
t_{KHTL}		9		11	ns
t_{KOH}	2		2		ns
t_{KP}	4		5		ns
t_{OHZ}		8		8	ns
t_{OLZ}	5		5		ns
t_{SP}	3		3		ns

Figure 5: Asynchronous WRITE#

Table 4: Asynchronous WRITE Timing Parameters

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
t_{AS}	0		0		ns
t_{AVH}	5		5		ns
t_{AVS}	10		10		ns
t_{AW}	70		85		ns
t_{BW}	70		85		ns
t_{CEW}	1	7.5	1	7.5	ns
t_{CW}	70		85		ns
t_{DH}	0		0		ns
t_{DW}	23		23		ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
t_{HZ}		8		8	ns
t_{LZ}	10		10		ns
t_{OW}	5		5		ns
t_{VP}	10		10		ns
t_{VPH}	10		10		ns
t_{VS}	70		85		ns
t_{WHZ}		8		8	ns
t_{WP}	46		55		ns
t_{WPH}	10		10		ns

Figure 6: Burst WRITE Operation


- Notes: 1. Non-default BCR settings for burst WRITE operation: Latency code two (three clocks); WAIT active LOW; WAIT asserted during delay.
 2. Clock rates below 50 MHz ($t_{CLK} > 20\text{ns}$) are allowed as long as t_{CSP} specifications are met.

Table 5: Burst WRITE Timing Parameters

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{CBPH}	5		5		ns
t_{CEM}		8		8	μs
t_{CEW}	1	7.5	1	7.5	ns
t_{CLK}	12.5	20	15	20	ns
t_{CSP}	4.5	20	5	20	ns
t_{HD}	2		2		ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{HZ}		8		8	ns
t_{KHL}		1.8		2.0	ns
t_{KHTL}		9		11	ns
t_{KP}	4		5		ns
t_{SP}	3		3		ns

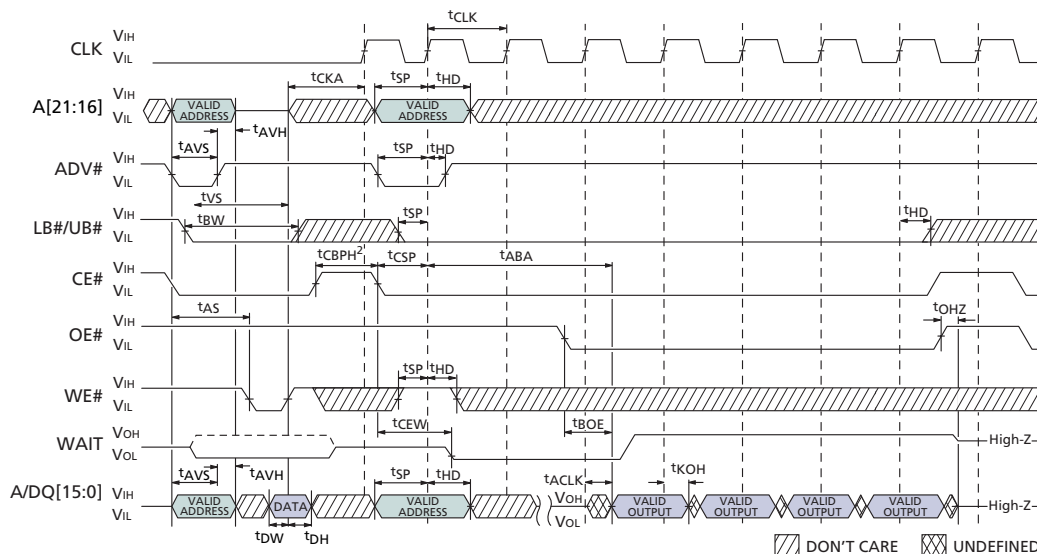
- Notes:
1. Non-default BCR settings for burst WRITE followed by burst READ: Latency code two (three clocks); WAIT active LOW; WAIT asserted during delay.
 2. When configured for synchronous mode (BCR[15] = 0), a refresh opportunity must be provided every t_{CEM} . A refresh opportunity is satisfied by either of the following two conditions: a) clocked CE# HIGH, or b) CE# HIGH for greater than 15ns. Note that CellularRAM Workgroup 1.0 specification requires CE# to be clocked HIGH to terminate the burst.
 3. Clock rates below 50 MHz ($t_{CLK} > 20ns$) are allowed as long as t_{CSP} specifications are met.

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
^t CBPH	5		5		ns
^t CLK	12.5	20	15	20	ns
^t CSP	4.5	20	5	20	ns

	-708		-706/-856		
Symbol	MIN	MAX	MIN	MAX	Units
^t HD	2		2		ns
^t SP	3		3		ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
^t ABA		46.5		56	ns
^t ACLK		9		11	ns
^t BOE		20		20	ns
^t CLK	12.5	20	15	20	ns
^t CSP	4.5	20	5	20	ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
^t HD	2		2		ns
^t KOH	2		2		ns
^t OHZ		8		8	ns
^t SP	3		3		ns

Figure 8: Asynchronous WRITE Followed by Burst READ


- Notes: 1. Non-default BCR settings for asynchronous WRITE followed by burst READ: Latency code two (three clocks); WAIT active LOW; WAIT asserted during delay.
2. When configured for synchronous mode (BCR[15] = 0), a refresh opportunity must be provided every t_{CEM} . A refresh opportunity is satisfied by either of the following two conditions: a) clocked CE# HIGH, or b) CE# HIGH for greater than 15ns. Note that CellularRAM Workgroup 1.0 specification requires CE# to be clocked HIGH to terminate the burst.
3. Clock rates below 50 MHz ($t_{CLK} > 20ns$) are allowed as long as t_{CSP} specifications are met.

Table 8: WRITE Timing Parameters – Async WRITE Followed by Burst READ

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
t_{AS}	0		0		ns
t_{AVH}	5		5		ns
t_{AVS}	10		10		ns
t_{AW}	70		85		ns
t_{BW}	70		85		ns
t_{CKA}	70		85		ns
t_{CVS}	10		10		ns
t_{CW}	70		85		ns
t_{DH}	0		0		ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
t_{DW}	20		23		ns
t_{VP}	10		10		ns
t_{VS}	70		85		ns
t_{WHZ}		8		8	ns
t_{WP}	46		55		ns
t_{WR}	0		0		ns

Table 9: READ Timing Parameters – Async WRITE Followed by Burst READ

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{ABA}		46.5		56	ns
t_{ACK}		9		11	ns
t_{BOE}		20		20	ns
t_{CBPH}	5		5		ns
t_{CEW}	1	7.5	1	7.5	ns
t_{CLK}	12.5	20	15	20	ns

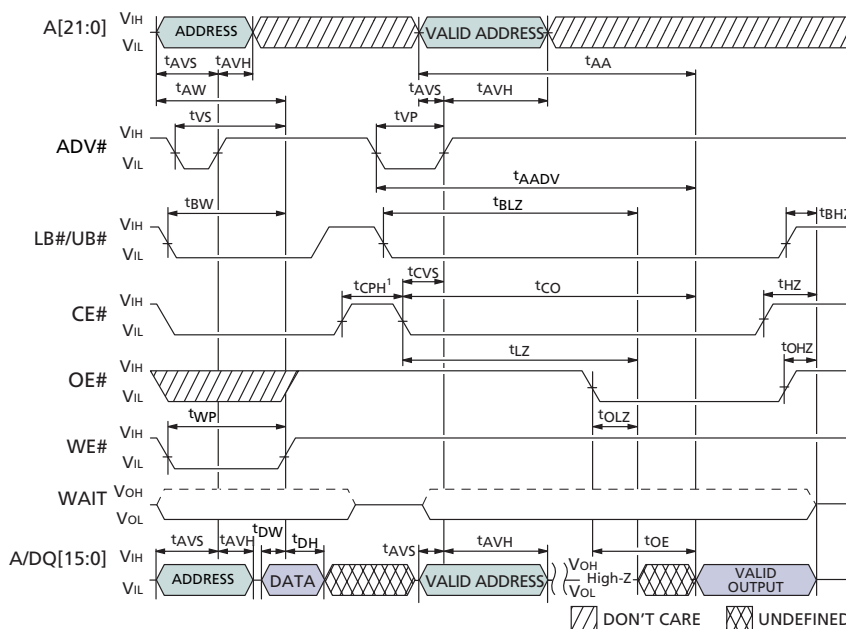
Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
t_{CSP}	4.5	20	5	20	ns
t_{HD}	2		2		ns
t_{KOH}	2		2		ns
t_{OHZ}		8		8	ns
t_{SP}	3		3		ns

Symbol	-708		-706/-856		Units
	MIN	MAX	MIN	MAX	
^t ABA		46.5		56	ns
^t ACLK		9		11	ns
^t BOE		20		20	ns
^t CBPH	5		5		ns
^t CEW	1	7.5	1	7.5	ns
^t CLK	12.5	20	15	20	ns
^t CSP	4.5	20	5	20	ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t AS	0		0		ns
^t AVH	5		5		ns
^t AVS	10		10		ns
^t AW	70		85		ns
^t BW	70		85		ns
^t CEW	1	7.5	1	7.5	ns
^t CW	70		85		ns
^t DH	0		0		ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t DW	23		23		ns
^t HZ		8		8	ns
^t VP	10		10		ns
^t VPH	10		10		ns
^t VS	70		85		ns
^t WP	46		55		ns
^t WPH	10		10		ns

Figure 10: Asynchronous WRITE Followed by Asynchronous READ



Notes: 1. When configured for synchronous mode (BCR[15] = 0), CE# must remain HIGH for at least 5ns (t_{CPH}) to schedule the appropriate internal refresh operation. Otherwise, t_{CPH} is only required after CE#-controlled WRITES.

Table 12: WRITE Timing Parameters – Async WRITE Followed by Async READ

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t AS	0		0		ns
^t AVH	5		5		ns
^t AVS	10		10		ns
^t AW	70		85		ns
^t BW	70		85		ns
^t CBPH	5		5		ns
^t CVS	10		10		ns
^t CW	70		85		ns
^t DH	0		0		ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t DW	23		23		ns
^t VP	10		10		ns
^t VPH	10		10		ns
^t VS	70		85		ns
^t WC	70		85		ns
^t WHZ		8		8	ns
^t WP	46		55		ns
^t WPH	10		10		ns
^t WR	0		0		ns

Table 13: READ Timing Parameters – Async WRITE Followed by Async READ

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t AA		70		85	ns
^t AADV		70		85	ns
^t AVH	5		5		ns
^t AVS	10		10		ns
^t BHZ		8		8	ns
^t BLZ	10		10		ns
^t CO		70		85	ns

Symbol	-70x		-856		Units
	MIN	MAX	MIN	MAX	
^t CVS	10		10		ns
^t HZ		8		8	ns
^t LZ	10		10		ns
^t OE		20		20	ns
^t OHZ		8		8	ns
^t OLZ	5		5		ns
^t VP	10		10		ns

Conclusion

By connecting A[15:0] to DQ[15:0] and implementing the timings shown in Figures 2–10, standard burst CellularRAM devices can be utilized with a x16 A/D multiplexed interface.



8000 S. Federal Way, P.O. Box 6, Boise, ID 83707-0006, Tel: 208-368-3900
prodmktg@micron.com www.micron.com Customer Comment Line: 800-932-4992
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